



COURSE SYLLABUS						
Course Code	INTE 301	Course Title	Systems Integration and Architecture 1		Course Credit	3
Course Description		This course introduces the concepts, methodologies, and technologies involved in system integration and architecture design. It focuses on the analysis, design, and implementation of integrated information systems that connect various hardware, software, and data resources within an enterprise. Students will explore architectural frameworks, integration models, middleware, API design, and enterprise service bus (ESB) applications. The course also emphasizes cloud computing, hybrid architectures, and the use of modern integration tools for system interoperability and scalability. By the end of the course, students are expected to design, document, and present an integrated system solution that demonstrates efficiency, security, and adaptability to evolving IT environments.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL
LEARNING OUTCOMES
(ILO)

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development;



	7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.
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PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain the fundamental concepts, frameworks, and models of system integration and architecture.								
	Apply integration techniques and tools in designing and developing integrated systems.								
	Implement cloud and hybrid architectures, ensuring security, interoperability, and adherence to best practices.								



	Demonstrate project management, collaboration, and evaluation skills through a final integration project.																
	Identify and compare different integration architectures and middleware solutions used in enterprise systems.																
	Design, test, and debug integrated systems using APIs, service orchestration, and data transformation tools.																
	Assess integration solutions based on performance, scalability, and maintainability.																
	Practice ethical, legal, and security standards in the design and implementation of system integrations.																



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Wee k	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignmen t to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.	Online meet for clarification of course requirements	Formative Assessment (Sharing of Learning)		X			
2	Define systems integration	Introduction to systems integration;	Lecture with examples and visuals.	Watch short video and write a short summary.	Oral Recitation.	Simple quiz on definitions.				



	and architecture and explain their importance in IT.	overview of system architecture in real-world industries.							
3	Identify the main components of system architecture. Explain how they work together.	Hardware, software, data, and network layers in a system.	Diagram explanation and group activity.	Search for a sample architecture online.	Group sharing.	Submit labeled diagrams.			
4	Describe different integration models and their purposes.	Point-to-Point, Hub-and-Spoke, ESB, API-based integration.	Discussion and sample diagram drawing.	Watch video on integration types.	Short quiz.	Reflection post comparing models.			
5	Explain middleware and APIs and how they connect applications.	Middleware concept, REST and SOAP API basics.	Demo using a sample API call (no coding).	Watch API demo video.	Participati on in demo discussion .	Short worksheet on "Where are APIs used?".	X	X	
6	Identify common integration	Issues on data mismatch, delay, and security	Case study sharing.	Read a short article on data issues.	Oral participati	Write "one challenge + one			



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	challenges and how to solve them.	risks.			on.	solution.”				
7	Recognize common enterprise architecture frameworks.	TOGAF, Zachman, and FEAF overview.	Lecture with slides.	Mini research on one framework.	Recitation.	Poster or infographic submission				

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QS WORLD UNIVERSITY RANKINGS

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

WURI

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21Rank 39Rank 50

QS STARS

★★★★★

EmployabilityTeachingOnline LearningAcademic DevelopmentInclusivenessInternationalization

★★★★★

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

★★★★★

Times Higher Education Impact Rankings 2024



8-9	• Create a simple integration plan or system flow.	• Steps in system design and integration planning.	• Workshop and guidance.	• Peer feedback session.	• Group plan presentation.	• Draft submission.				
MIDTERM WRITTEN EXAMINATION										
10	• Understand what cloud computing is and its services.	• IaaS, PaaS, SaaS; public and private cloud examples.	• Lecture-demo on cloud tools (Google Cloud).	• Watch tutorials and answer guide questions.	• Short quiz.	• Reflection on cloud use in daily life.				
11	• Differentiate between on-premise, cloud, and hybrid systems.	• Comparison of traditional vs cloud vs hybrid setups.	• Discussion and diagram comparison.	• Case reading: "How companies use hybrid systems."	• Recitation.	• Comparison table.				
12	• Explain the idea of Service-Oriented	• How web services connect apps; SOAP and REST.	• Lecture and simple web service demo.	• Read and summarize SOA basics.	• Seatwork.	• Quiz.				



	Architectu re (SOA).								
13	Understan d how legacy systems connect to cloud services.	Hybrid integration and middleware tools.	Instructor demo on hybrid setup.	Watch a short hybrid integration video.	Discussion questions.	Short reflection paper.			
14	Recognize security practices for integrated systems.	Authentication, authorization, and encryption.	Lecture with real-life examples.	Read a short article on data privacy.	Oral recitation.	5-item online quiz.			
15	Start developin g a simple integratio n project (mock).	Basic integration workflow using cloud or API tools.	Group consultation.	Online collaboration.	Progress report.	Screenshot of group work.			
16	Test and document the group project.	Testing methods and simple documentation.	Testing session and peer review.	Write a documentation draft.	Peer feedback.	Documentat ion upload.			
17-	Present	Final project	Group	Peer feedback.	Oral	Peer			



18	and explain the project output.	presentation and peer evaluation	presentation .		defense.	evaluation form.				
	FINAL WRITTEN EXAMINATION									



		- Resources and Applying Strategies in Teaching Prose - Competency 4: Recognizing Assessment in Teaching Prose	- achievement which are consistent with the selected competencies - Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: Microsoft. (n.d.). <i>Integration architecture design</i>. Microsoft Learn. Retrieved from https://learn.microsoft.com/en-us/azure/architecture/integration/integration-start-here Microsoft. (n.d.). <i>Cloud design patterns</i>. Microsoft Learn. Retrieved from https://learn.microsoft.com/en-us/azure/architecture/patterns/ Microsoft. (n.d.). <i>Basic enterprise integration on Azure</i>. Microsoft Learn. Retrieved from https://learn.microsoft.com/en-us/azure/architecture/reference-architectures/enterprise-integration/basic-enterprise-integration Google Cloud. (2024). <i>Google Cloud well-architected framework</i>. Retrieved from https://cloud.google.com/architecture/framework Amazon Web Services. (n.d.). <i>AWS well-architected framework</i>. Retrieved from https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html										



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only.



	Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	30%	Final Examinations	
	100%		30%
			100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM
- Students are expected to actively participate in class discussions and online activities, complete quizzes and hands-on exercises, and submit individual or group outputs related to system design and integration. - A midterm exam, final exam, and a final integration project demonstrating the	Final Grade = (Midterm Grade + Final Term Grade) / 2



application of cloud and hybrid architecture concepts are required to successfully complete the course.			
CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section: BSIT 3-1		Name of Faculty: MR. Louie Carl R. Mandapat	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:	

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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MR. LOUIE CARL R. MANDAPAT Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	 DR. RHYAN V. MOLINAR Head of Academic Programs	DR. ARMANDO A. TORRES Vice President for Campuses
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for INNOVATION
Rank 21
Rank 39
Rank 50

**STARS**

**Impact Rankings 2024**



- Employability
- Teaching
- Online Learning
- Academic Development
- Internationalization



- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)
- Inclusiveness